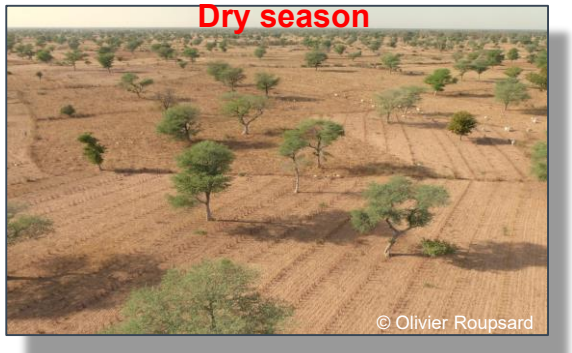
Modelling CO_2 and N_2O emissions from a tropical semi-arid parkland cultivated with groundnut and millet

Yélognissè Agbohessou*, *Seydina Ba, Fabien Ferchaud, Joël Léonard, Sidy Sow, Frédéric Bouvery, Maxime Duthoit, Claire Delon, Rémi Vezy, Gatien N. Falconnier, Eric Mougin, Daouda Ngom, Simon Taugourdeau, Olivier Roupsard*

"Faidherbia-Flux" Web site :
https://lped.info/wikiObsSN/?Faidherbia-Flux





Contact:
ayulrich@yahoo.fr



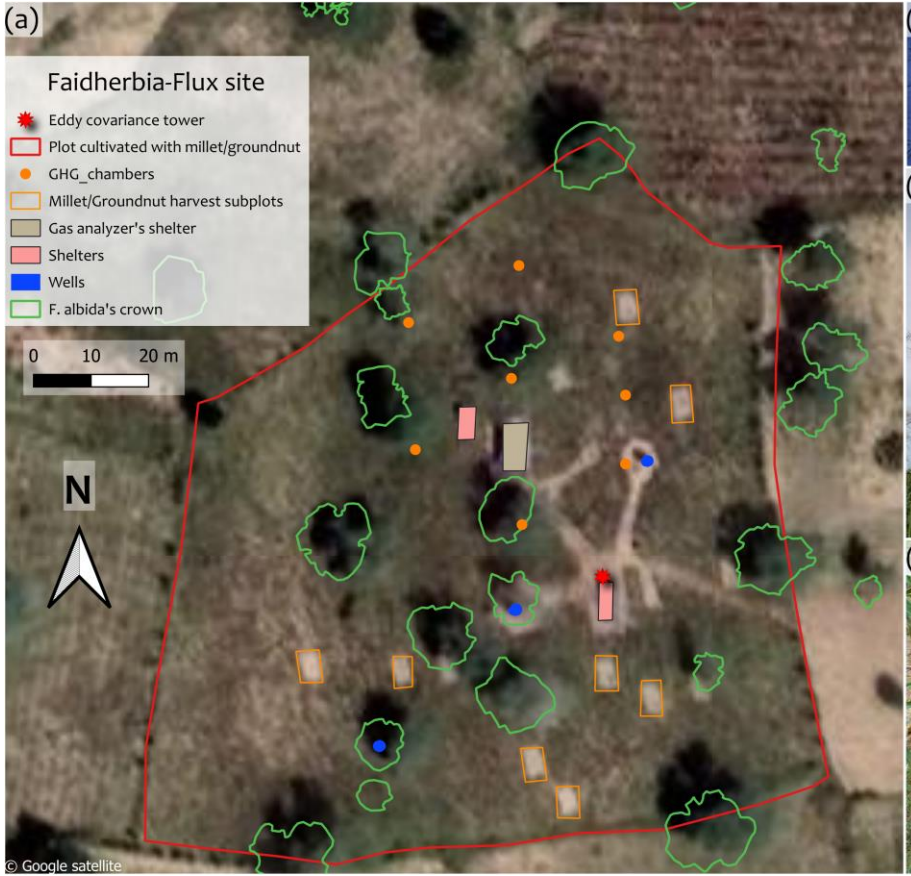
Objectives

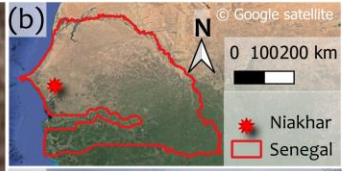
(i) Assess the potential of the STICS model to simulate, soil moisture, soil temperature, N_2O emissions, and heterotrophic respiration;

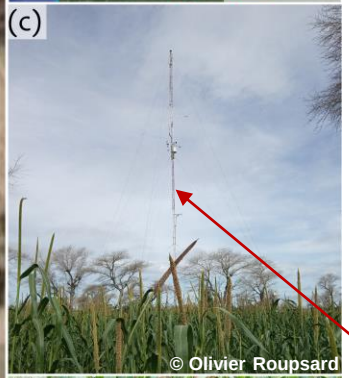
(ii) Simulate the autotrophic respiration of millet and groundnut using an autotrophic respiration module; and

(iii) Evaluate the potential of the STICS model and the autotrophic respiration module to simulate measured total CO_2 emissions.

Study area: the "Faidherbia-Flux" site









Sudano-Sahelian climate (2018-2022) **:**
Mean annual precipitation = 574 mm
Average annual air temperature = 27.62 °C

Sol : Sandy soils (82.1 % of sand, 8.9 % of clay). pH_{H2O} = 6.52

Vegetation : Trees (*Faidherbia albida*) & Crops (millet and groundnut in rotation since 2018)

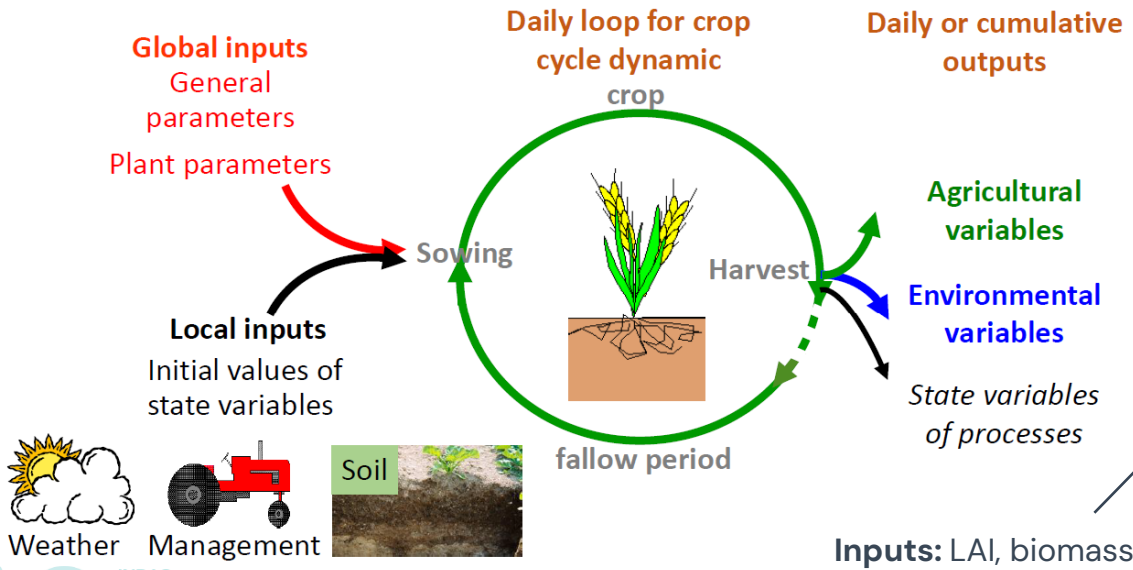
Eddy Covariance flow tower (20 m high)

GHG measurement chamber
(Seydina Ba's thesis)
2021 : crops in the chambers
2022 : no crops in the chambers

Materials and methods: models description

STICS

Simulation of water balance, heterotrophic respiration and soil N_2O emissions



+

Autotrophic respiration module (DynACof)

Inputs: LAI, biomass of crop organs simulated by STICS, global radiation and air temperature.

Outputs: autotrophic respiration of millet and groundnut

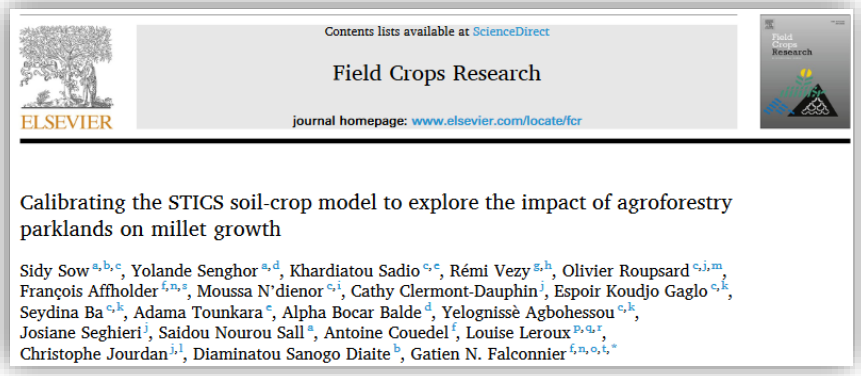
Materials and methods: description of the experimental set-up used for simulation with STICS on the "Faidherbia-Flux" site

Year	Crop	Variety	Density (plant/m²)	Stat USM	Sowing date	Harvest date	End USM	Soil GHG flux measurement	USM name*
2018	Millet	Souna	1.23	19/06/2018	19/07/2018	31/10/2018	31/12/2018	No	Nkr_millet_sun_2018
2019	Groundnut	Fleur 11	10	01/11/2018	27/07/2019	07/11/2019	31/12/2019	No	Nkr_grdnut_sun_2019
2020	Millet	Souna	1.23	08/11/2019	06/07/2020	30/10/2020	31/12/2020	No	Nkr_millet_sun_2020
2021	Groundnut	Fleur 11	10	31/10/2020	25/06/2021	13/10/2021	31/12/2021	Yes	Nkr_grdnut_sun_2021
2022	Millet	Souna	1.23	14/10/2021	10/06/2022	29/09/2022	31/12/2022	Yes	Nkr_millet_sun_2022

USM = Simulation unit*

Materials and methods: simulation strategy (STICS)

STICS calibration for millet



Contents lists available at ScienceDirect

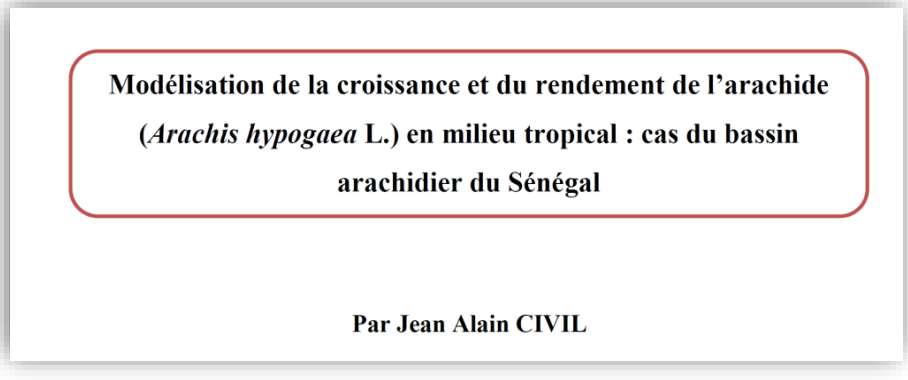
Field Crops Research

journal homepage: www.elsevier.com/locate/fcr

Calibrating the STICS soil-crop model to explore the impact of agroforestry parklands on millet growth

Sidy Sow [a,b,c], Yolande Senghor [a,d], Khardiatou Sadio [c,e], Rémi Vezy [g,h], Olivier Roupsard [c,j,m], François Affholder [f,n,s], Moussa N'dienor [c,i], Cathy Clermont-Dauphin [j], Espoir Koudjo Gaglo [c,k], Seydina Ba [c,k], Adama Tounkara [e], Alpha Bocar Balde [d], Yelognissè Agbohessou [c,k], Josiane Seghieri [j], Saidou Nourou Sall [a], Antoine Couedel [f], Louise Leroux [p,q,r], Christophe Jourdan [j,l], Diaminatou Sanogo Diaite [b], Gatien N. Falconnier [f,n,o,t,*]

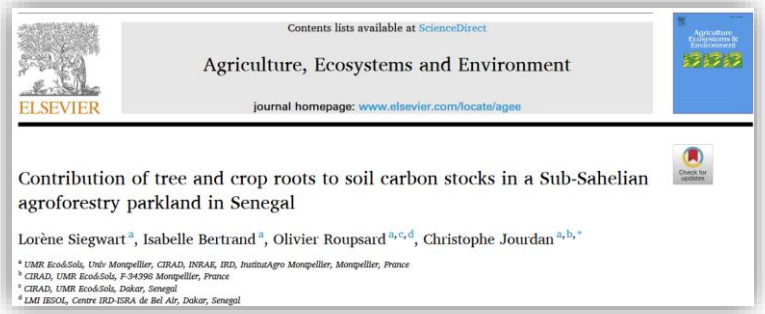
STICS calibration for groundnuts



Modélisation de la croissance et du rendement de l'arachide (*Arachis hypogaea* L.) en milieu tropical : cas du bassin arachidier du Sénégal

Par Jean Alain CIVIL

Calibrated root parameters for millet and groundnut



Contents lists available at ScienceDirect

Agriculture, Ecosystems and Environment

journal homepage: www.elsevier.com/locate/agee

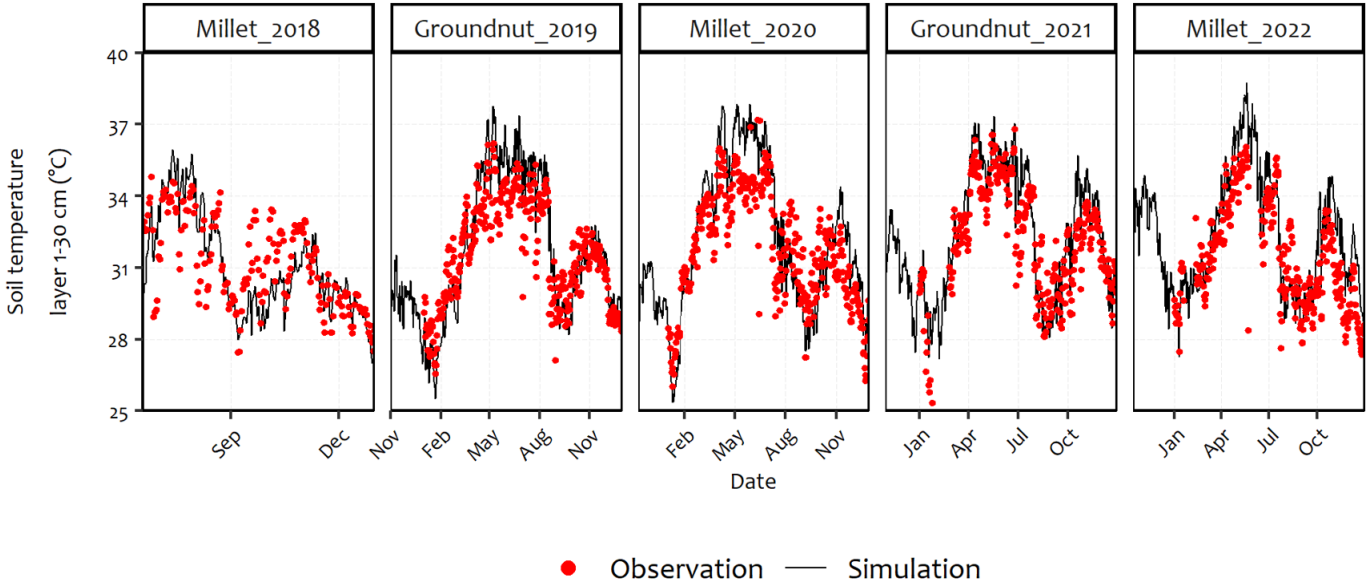
Contribution of tree and crop roots to soil carbon stocks in a Sub-Sahelian agroforestry parkland in Senegal

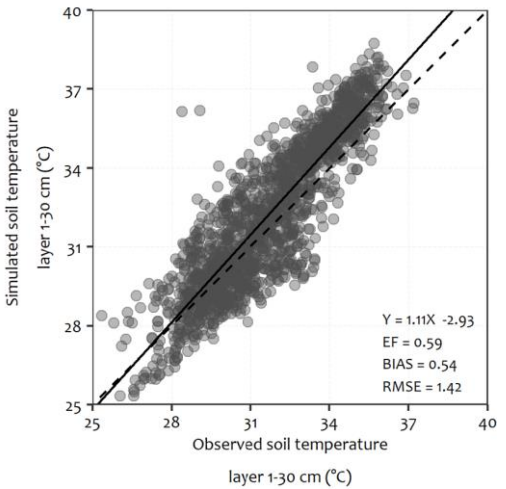
Lorène Siegwart [a], Isabelle Bertrand [a], Olivier Roupsard [a,c,d], Christophe Jourdan [a,b,*]

[a] UMR Eco&Sols, Univ Montpellier, CIRAD, INRAE, IRD, InstitutAgro Montpellier, Montpellier, France
[b] CIRAD, UMR Eco&Sols, F-34398 Montpellier, France
[c] CIRAD, UMR Eco&Sols, Dakar, Senegal
[d] LMI IESOL, Centre IRD-ISRA de Bel Air, Dakar, Senegal

- The model is initialized each year the day after the previous crop is harvested, adding the residues of the previous crop left on the ground (underground crop biomass) and the biomass of weeds on the ground.

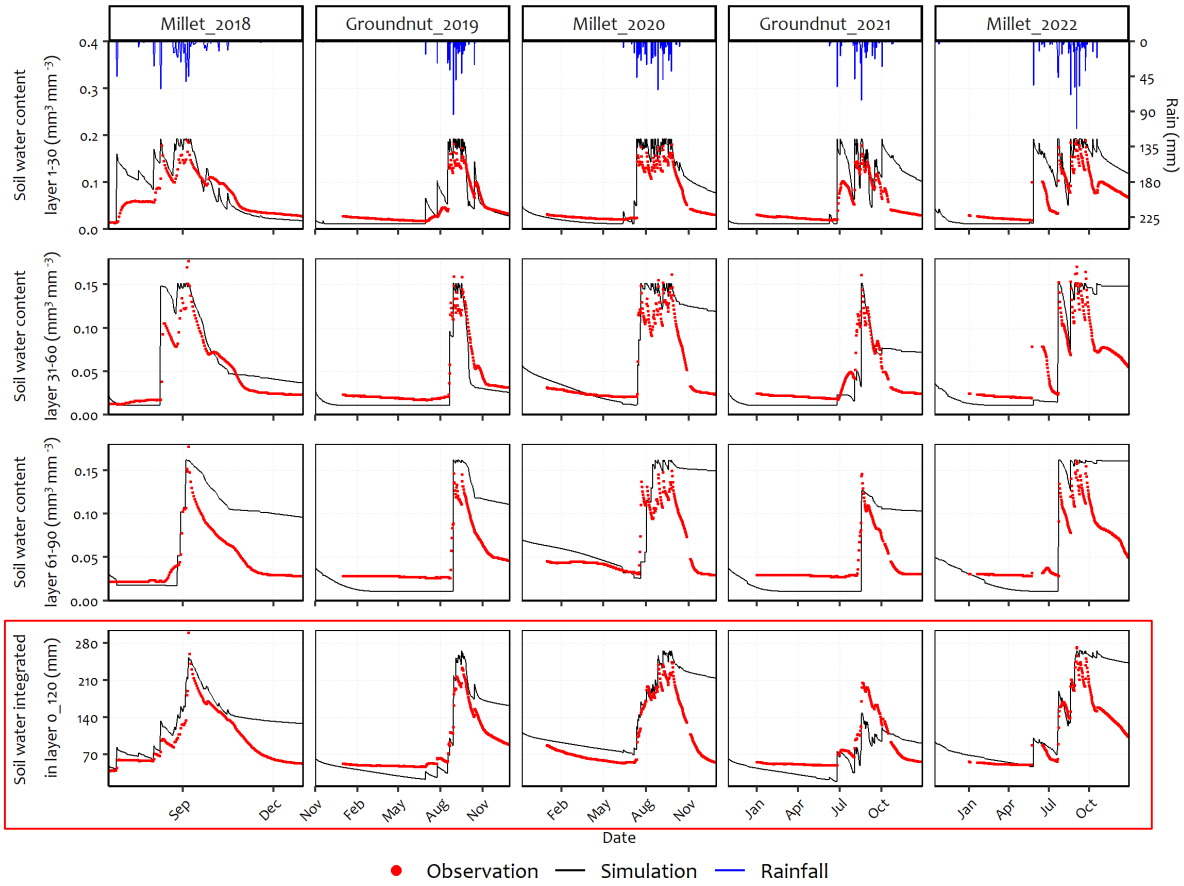
Result: soil temperature





- the STICS model reproduces well the temporal dynamics of soil temperature

Result: soil water content



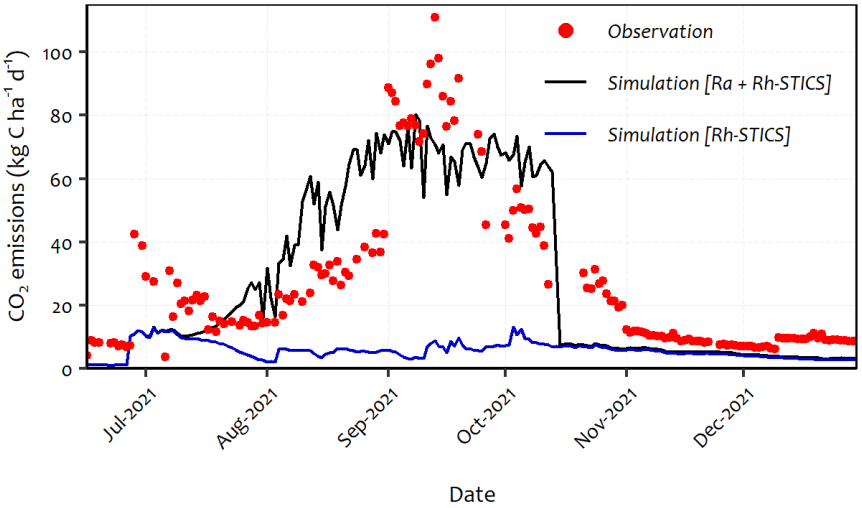
- STICS represents soil water content fairly well during the first part of the rainy season,
- STICS overestimates soil water content at the end of the crop cycle
- inaccuracies in the formalism governing soil evaporation????

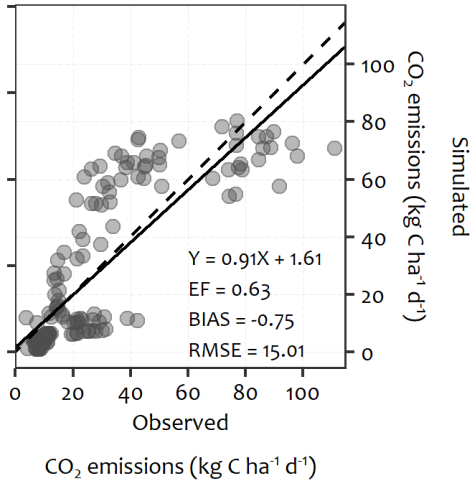
Result: CO_2 emissions (2021)

2021



Groundnut plot, GHG flux measurement by Seydina Ba





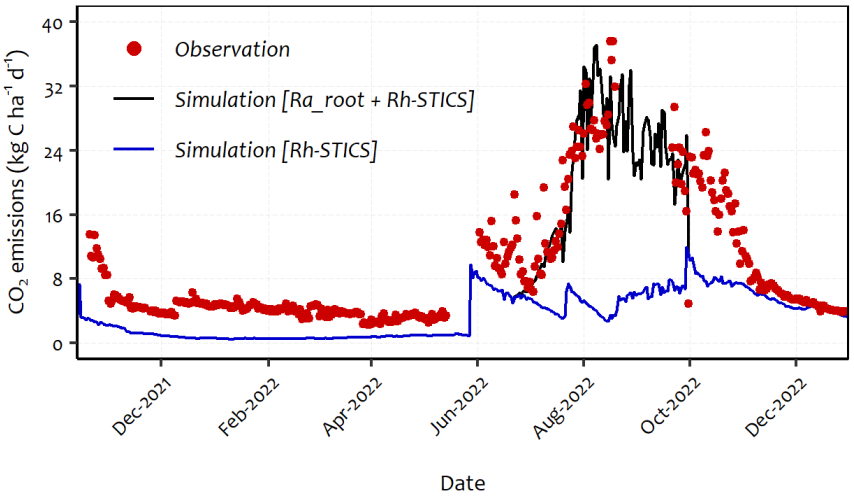
- The models underestimate the peak in total emissions measured during the rainy season.
- The temporal dynamics of measured total CO_2 emissions are not well represented by the models.

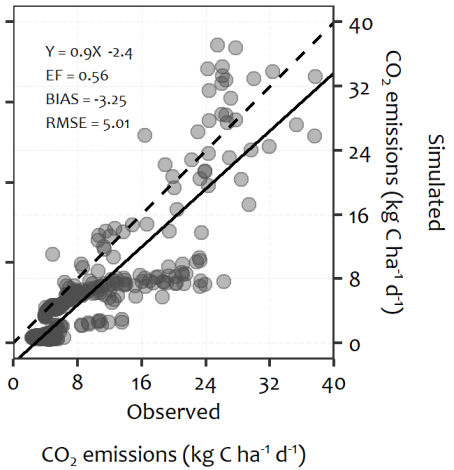
Result: CO_2 emissions (2022)

2022



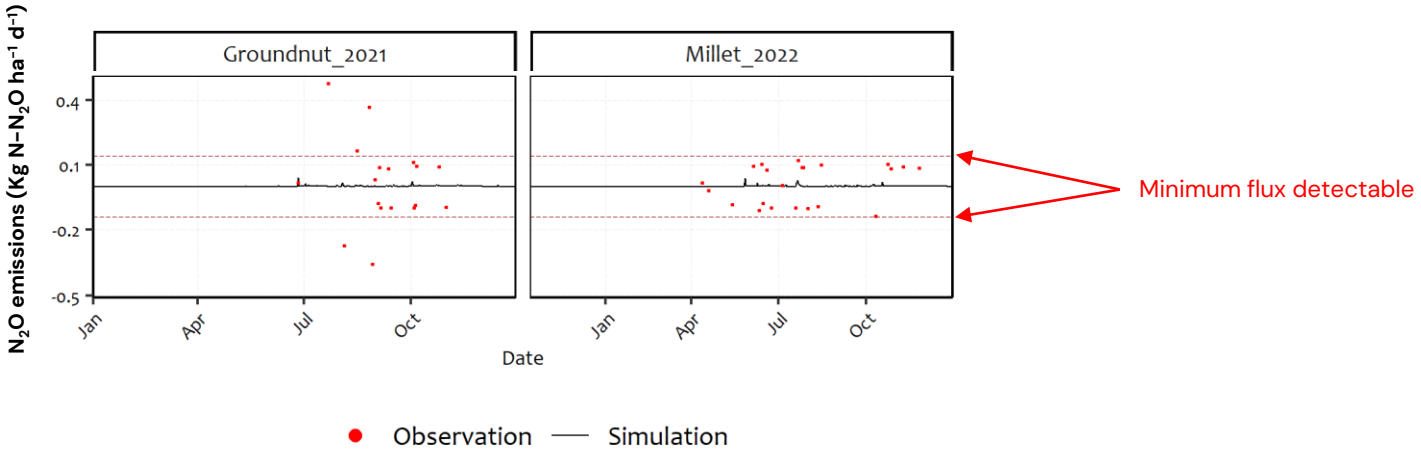
Millet plot, GHG flux measurement by Seydina Ba





- The models satisfactorily reproduce the gradual increase in total CO_2 emissions measured during the millet growing season, but underestimate total emissions measured during the dry season and post-harvest.

Result: N_2O emissions



- STICS does not currently include N_2O absorption by soil
- The STICS model does not satisfactorily reproduce measured N_2O emissions, which are mainly below the detection limit.

Conclusions and perspectives

- STICS: very satisfactory results for soil temperature, acceptable for soil moisture at the start of the rainy season and less accurate at the end of the rainy season, but promising for CO_2 emissions.
- The STICS model does not satisfactorily reproduce measured N_2O emissions, which are mainly below the detection limit.
- **Next step?**

➢ Modifications to the STICS formalism to improve simulation of minimum humidity during the dry season.

➢ Development of specific modules for calculating N_2O uptake, CO_2 fluxes between plant and atmosphere, and CH_4 emissions and uptakes, in order to broaden the application possibilities of the STICS model and provide a complete balance of GHG emissions.



Thank you for your attention!

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